ECEN 714: DIGITAL INTEGRATED CIRCUIT DESIGN
LAB REPORT - 8

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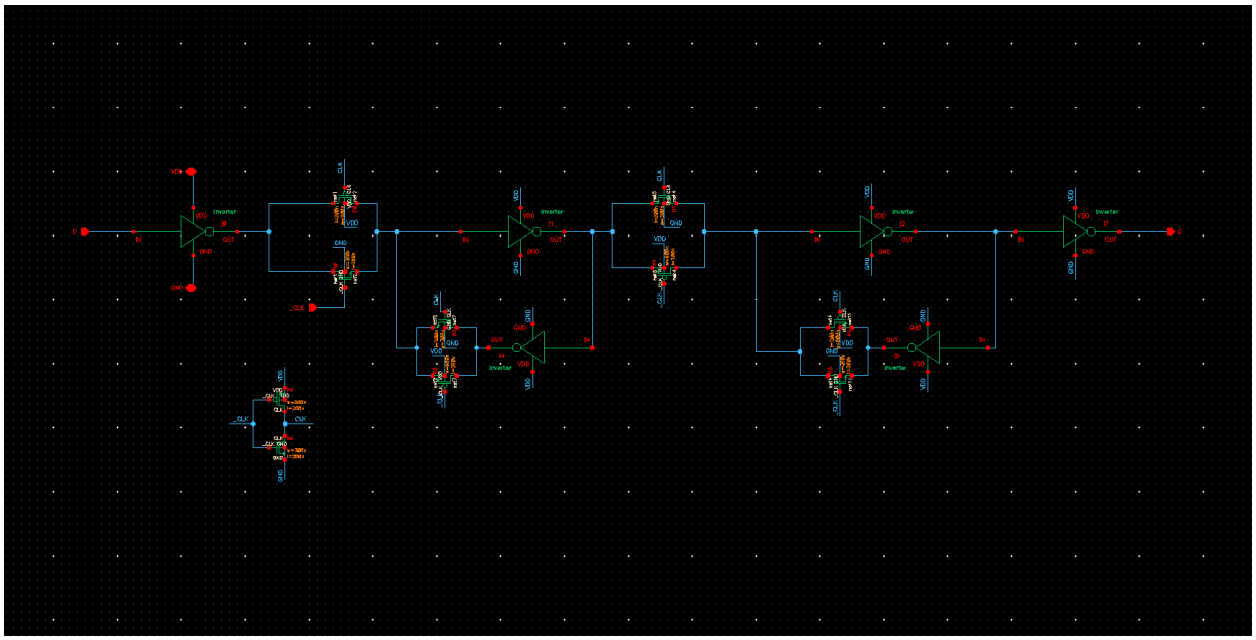
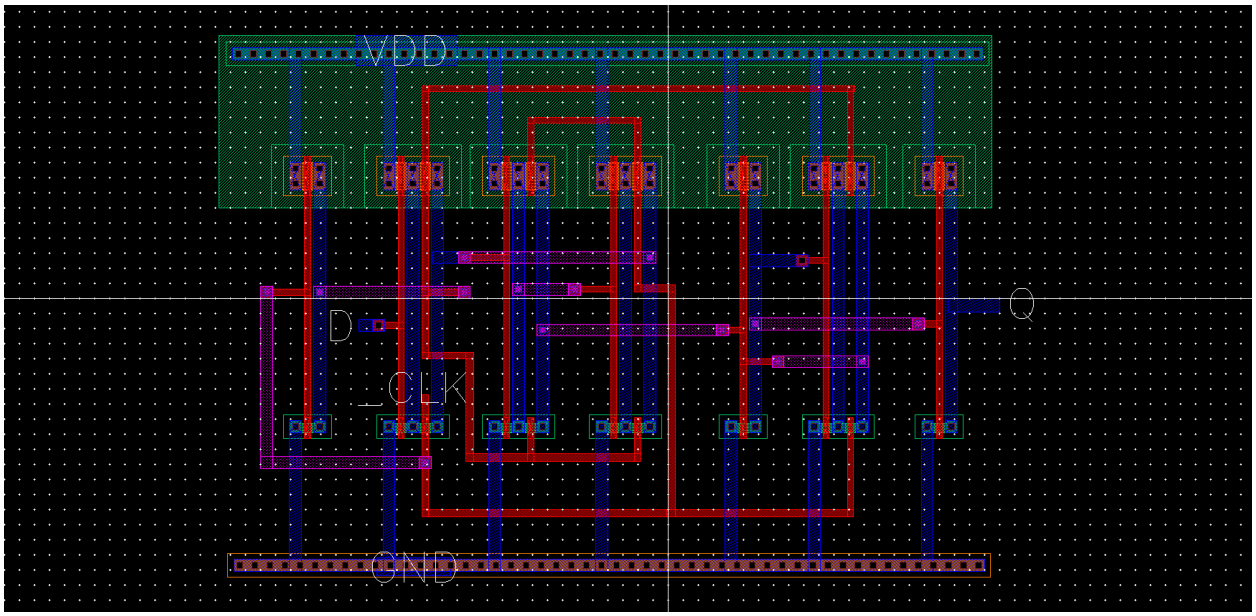
D-Flip Flop

1. Schematic:



The previous design of the flip flop was changed in order to maintain only one clock input that is _CLK.

2. Layout:

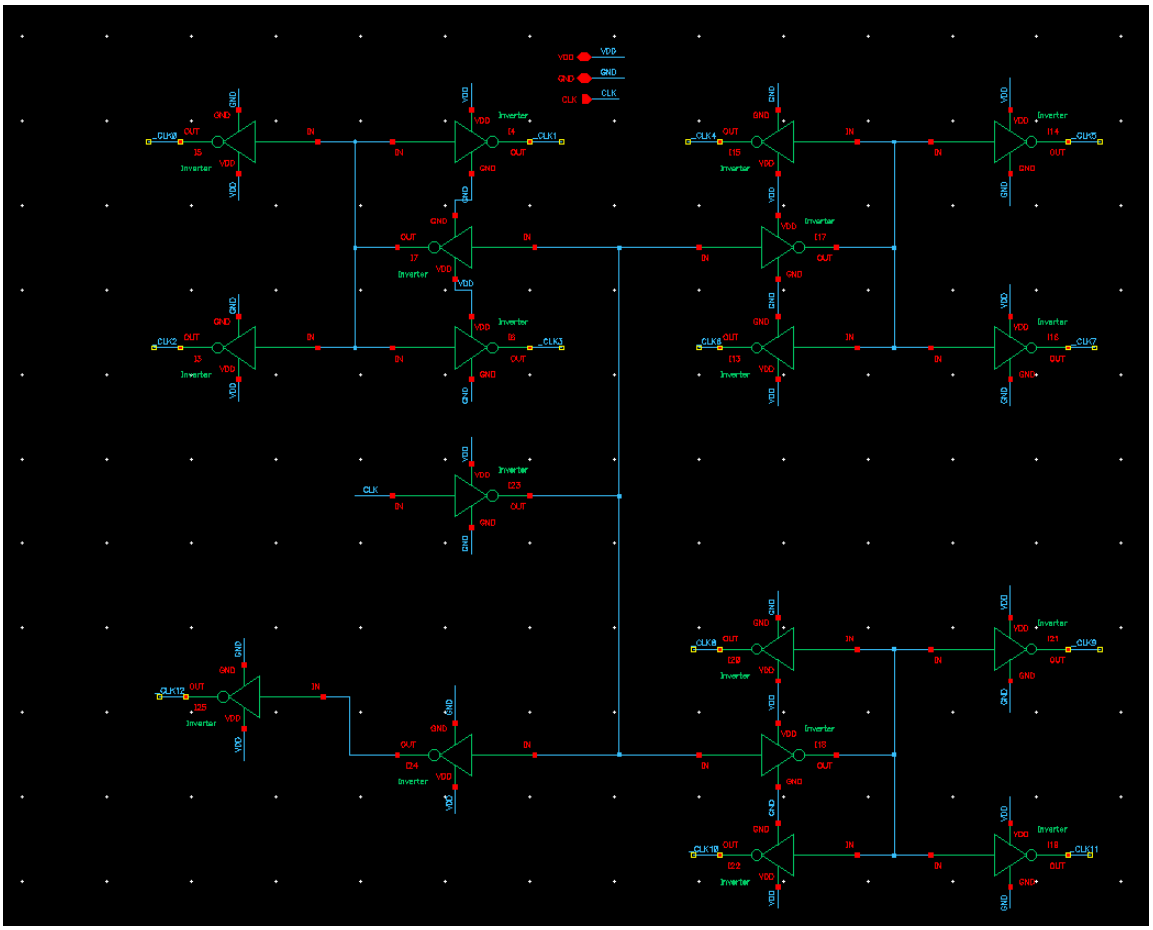


3. LVS Report:

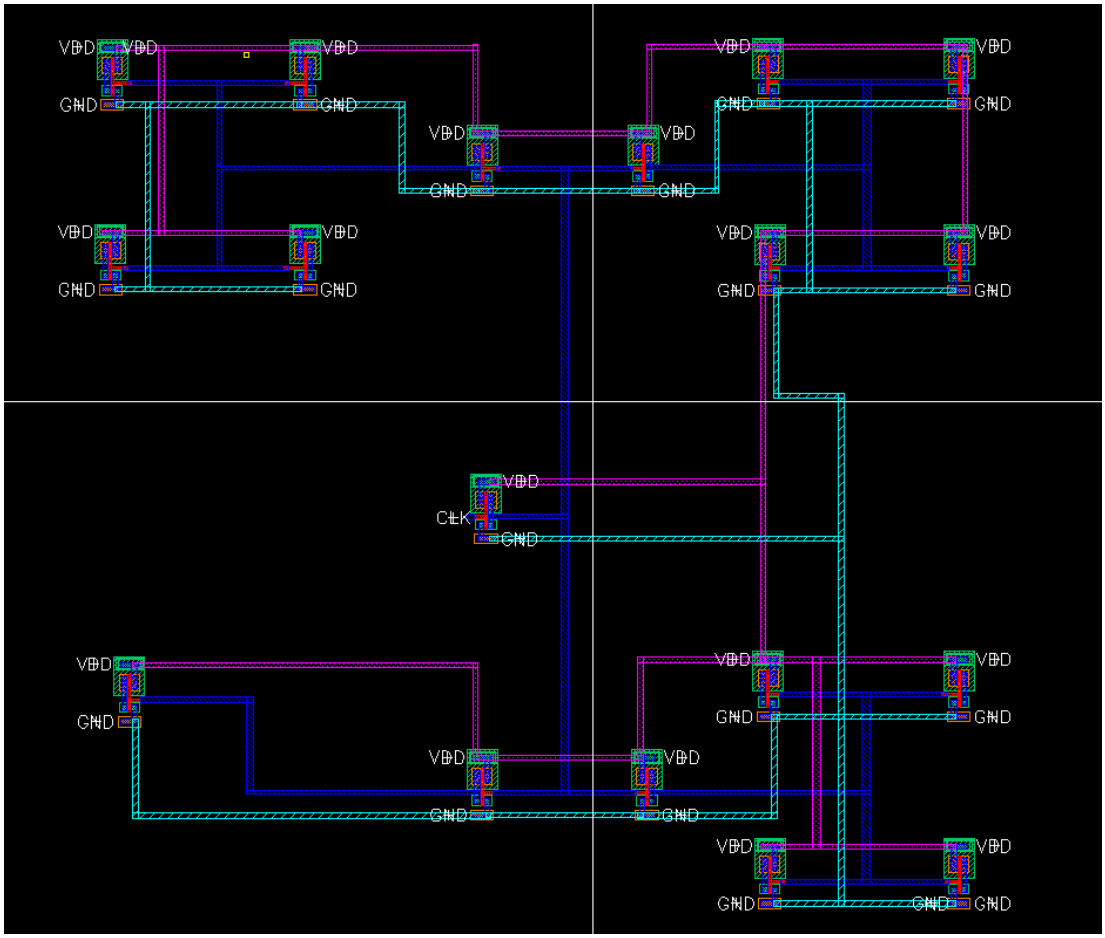
```
@(#)$CDS: LVS version 6.1.8-64b 09/03/2024 18:56 (sjfhw316) $

Command line: /opt/coe/cadence/IC618/tools.lnx86/dfII/bin/64bit/LVS -dir /home/grads/y/yashas1729/LVS -l -s -t /home/grads/y/yashas17
Like matching is enabled.
Net swapping is enabled.
Using terminal names as correspondence points.
Compiling Diva LVS rules...

    Net-list summary for /home/grads/y/yashas1729/LVS/layout/netlist
       count
        13              nets
        5               terminals
        11              pmos
        11              nmos

    Net-list summary for /home/grads/y/yashas1729/LVS/schematic/netlist
       count
        13              nets
        5               terminals
        11              pmos
        11              nmos


    Terminal correspondence points
    N11       N6        D
    N8        N10       GND
    N9        N8        Q
    N12       N9        VDD
    N10       N12       _CLK

Devices in the netlist but not in the rules:
        pcapacitor
Devices in the rules but not in the netlist:
        cap nfet pfet nmos4 pmos4

The net-lists match.

                              layout  schematic
                                 instances
        un-matched               0       0
        rewired                  0       0
        size errors              0       0
        pruned                   0       0
        active                   22      22
        total                    22      22
```

H-Clock Tree

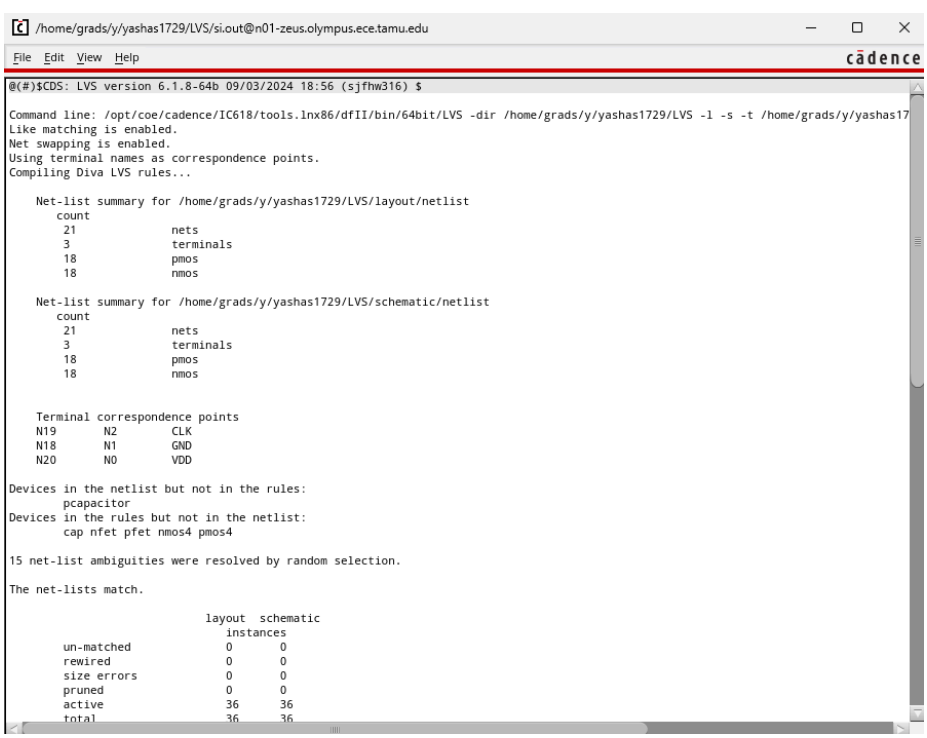
1. Schematic:



2. Layout:



3. LVS:



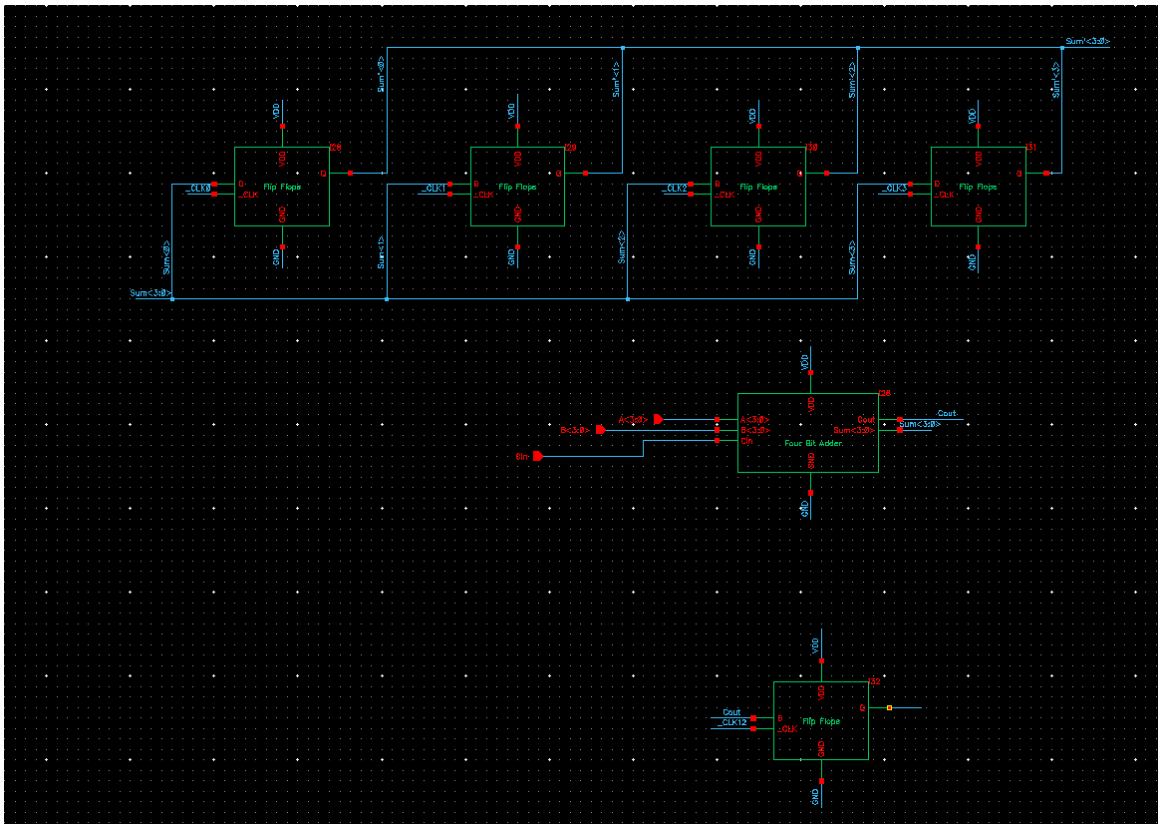
/home/grads/y/yashas1729/LVS/si.out@n01-zeus.olympus.ece.tamu.edu

File Edit View Help

```
@(#)$CDS: LVS version 6.1.8-64b 09/03/2024 18:56 (sjfhw316) $

Command line: /opt/coe/cadence/IC618/tools.lnx86/dfII/bin/64bit/LVS -dir /home/grads/y/yashas1729/LVS -l -s -t /home/grads/y/yashas17
Like matching is enabled.
Net swapping is enabled.
Using terminal names as correspondence points.
Compiling Diva LVS rules...

    Net-list summary for /home/grads/y/yashas1729/LVS/layout/netlist
       count
        21              nets
        3               terminals
        18              pmos
        18              nmos

    Net-list summary for /home/grads/y/yashas1729/LVS/schematic/netlist
       count
        21              nets
        3               terminals
        18              pmos
        18              nmos


    Terminal correspondence points
    N19       N2        CLK
    N18       N1        GND
    N20       N0        VDD

Devices in the netlist but not in the rules:
        pcapacitor
Devices in the rules but not in the netlist:
        cap nfet pfet nmos4 pmos4

15 net-list ambiguities were resolved by random selection.

The net-lists match.

                            layout  schematic
                               instances
        un-matched              0       0
        rewired                 0       0
        size errors             0       0
        pruned                  0       0
        active                  36      36
        total                   36      36
```

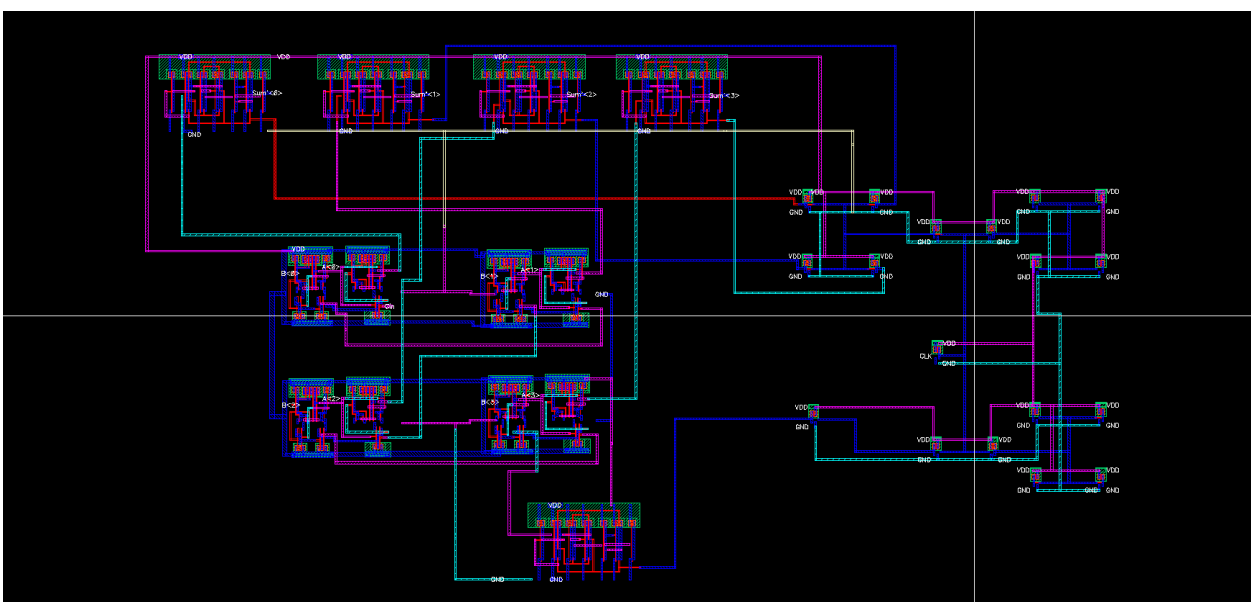
H-Clock Tree with 4-bit Adder

1. Schematic:



As you can see, the 4 bit adder along with the 4 bit register at the top with the D-Flip Flop is implemented on the left side with the H clock tree.

2. Layout:



3. LVS:

/home/grads/y/yashas1729/LVS/si.out@n01-zeus.olympus.ece.tamu.edu

File Edit View Help

```
@(#)$CDS: LVS version 6.1.8-64b 09/03/2024 18:56 (sjfhw316) $

Command line: /opt/coe/cadence/IC618/tools.lnx86/dfII/bin/64bit/LVS -dir /home/grads/y/yashas1729/LVS -l -s -t /home/grads/y/yashas1729/LVS/layou
Like matching is enabled.
Net swapping is enabled.
Using terminal names as correspondence points.
Compiling Diva LVS rules...

    Net-list summary for /home/grads/y/yashas1729/LVS/layout/netlist
       count
        155             nets
        16              terminals
        145             pmos
        145             nmos

    Net-list summary for /home/grads/y/yashas1729/LVS/schematic/netlist
       count
        155             nets
        16              terminals
        145             pmos
        145             nmos


    Terminal correspondence points
    N146      N24       A<0>
    N143      N23       A<1>
    N140      N22       A<2>
    N139      N21       A<3>
    N154      N28       B<0>
    N152      N27       B<1>
    N151      N26       B<2>
    N149      N25       B<3>
    N145      N2        CLK
    N150      N29       Cin
    N142      N1        GND
    N148      N35       Sum'<0>
    N147      N39       Sum'<1>
    N144      N36       Sum'<2>
    N141      N37       Sum'<3>
    N153      N0        VDD

Devices in the netlist but not in the rules:
        pcapacitor
Devices in the rules but not in the netlist:
        cap nfet pfet nmos4 pmos4

10 net-list ambiguities were resolved by random selection.

The net-lists match.

                              layout  schematic
                                 instances
        un-matched               0       0
        rewired                  0       0
        size errors              0       0
```

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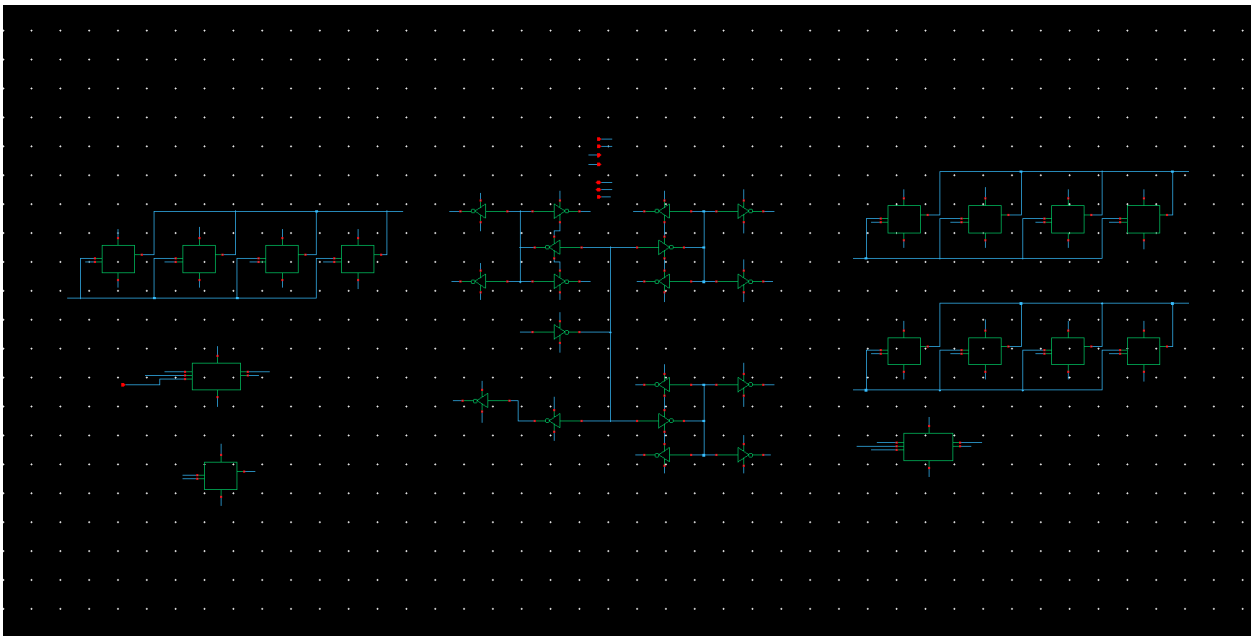
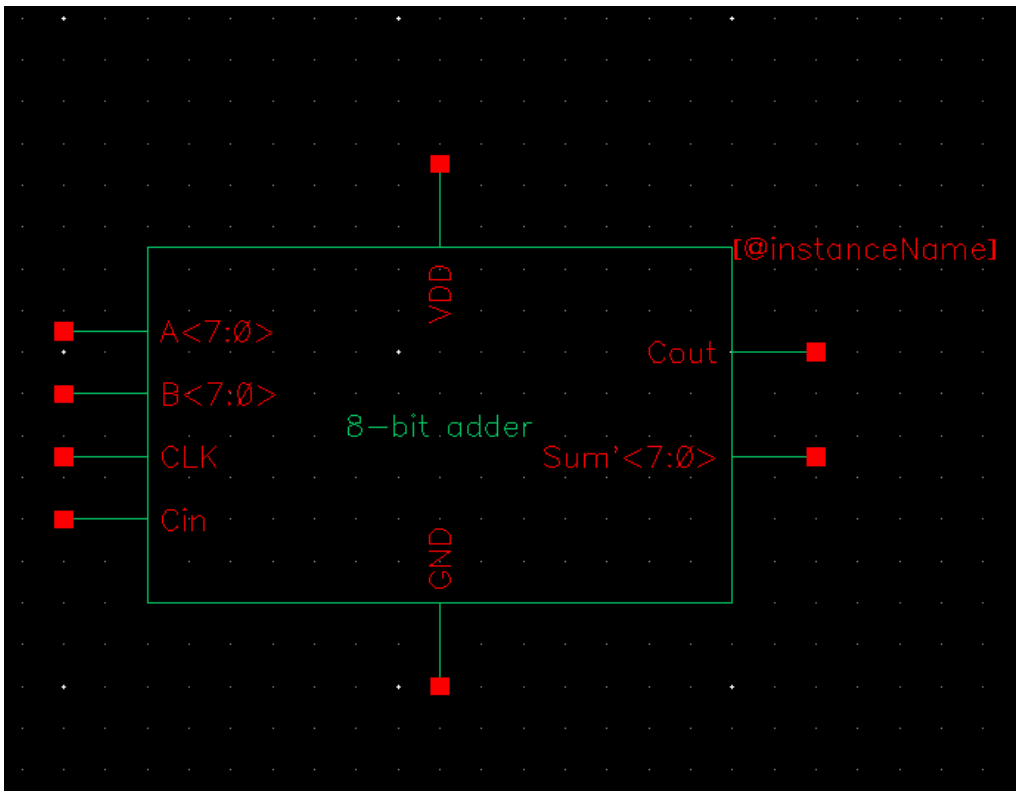
8-Bit Adder

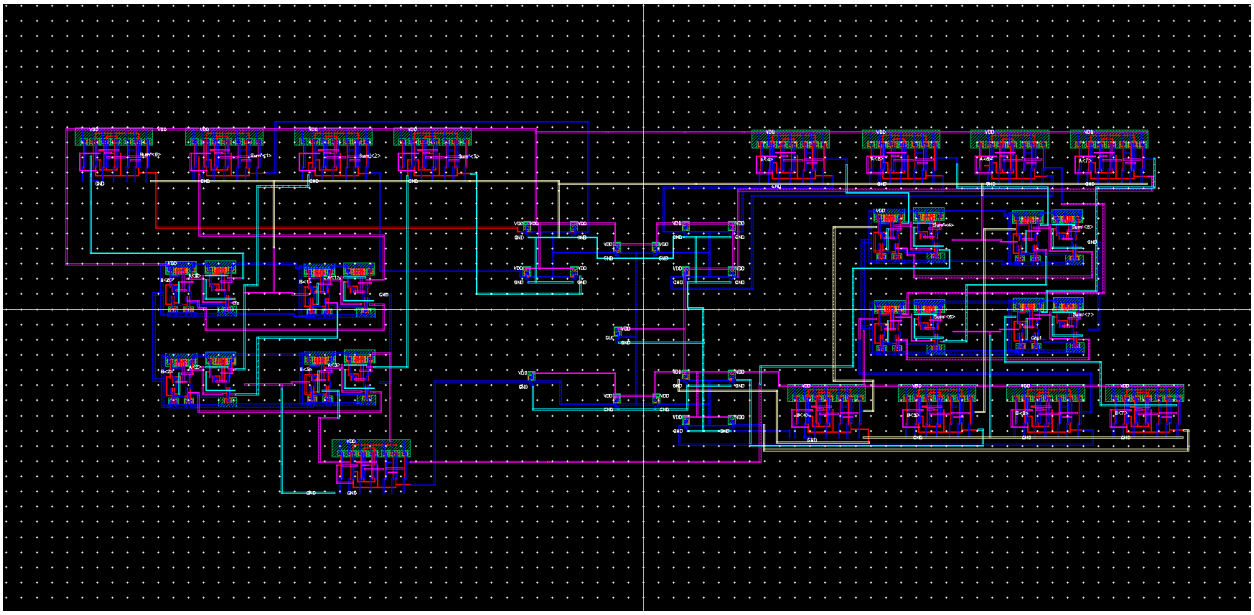
1. Schematic:



2. Symbol:



3. Layout:



4. DRC:

Virtuoso® 6.1.8-64b - Log: /home/grads/y/yashas1729/CDS.log@n02-hera.olympus.engr.tamu.edu

File Tools Options Help

```
Inclusion limit is set to 1000.
Parsing drcExtractRules of "/opt/coe/ncsu/ncsu-cdk-1.6.0.beta/lib/NCSU_TechLib_tsmc02/divaDRC.rul"...
Optimizing rules...
removing unused task: nwellResEdge = geomGetEdge(nwellRes coincident nwell)
removing unused task: polyResEdge = geomGetEdge(polyRes coincident poly)
removing unused task: sGateWidthCheckEdge = geomGetEdge(sGateWidthCheck)
removing unused task: metalcapEdge = geomGetEdge(metalcap)
removing unused task: padEdge = geomGetEdge(pad)
removing unused task: ccEdge = geomGetEdge(cc)
removing unused task: gselectEdge = geomGetEdge(gselect)
removing unused task: gwellEdge = geomGetEdge(gwell)
removing unused task: nwellRes = geomButting(geomAnd(res_id nwell) nBulk (keep == 2))
removing unused task: polyRes = geomButting(geomAndNot(geomAnd(res_id poly) polySRes) fieldPoly (keep == 2))
removing unused task: sGateWidthCheck = geomSize(geomSize(geomAnd(Gate sblock) -2.9) 2.9)
removing unused task: NwPdiode = geomAnd(dio_id geomOutside(nwell pNotOhmic))
removing unused task: PNdiode = geomAnd(dio_id geomOutside(pNotOhmic poly))
removing unused task: NPdiode = geomAnd(dio_id geomOutside(nNotOhmic poly))
removing unused task: m6m5Cap = geomAnd(geomAnd(metal5 metal6) cap_id)
removing unused task: m5m4Cap = geomAnd(geomAnd(metal4 metal5) cap_id)
removing unused task: m4m3Cap = geomAnd(geomAnd(metal3 metal4) cap_id)
removing unused task: m3m2Cap = geomAnd(geomAnd(metal2 metal3) cap_id)
removing unused task: m2m1Cap = geomAnd(geomAnd(metal1 metal2) cap_id)
removing unused task: m1sCap = geomAnd(geomAndNot(metal1 poly) cap_id)
removing unused task: m1pCap = geomAnd(geomAnd(poly metal1) cap_id)
removing unused task: pChannelTran = geomAndNot(pChannelTran hvpChannelTran)
removing unused task: hvpChannelTran = geomAnd(pChannelTran tactive)
removing unused task: nChannelTran = geomAndNot(nChannelTran hvnChannelTran)
removing unused task: hvnChannelTran = geomAnd(nChannelTran tactive)
removing unused task: pChannelCap = geomButting(pChannel pDiff (keep == 1))
removing unused task: nChannelCap = geomButting(nChannel nDiff (keep == 1))
removing unused task: pChannelTran = geomButting(pChannel pDiff (keep == 2))
removing unused task: nChannelTran = geomButting(nChannel nDiff (keep == 2))
removing unused task: Space = geomNot(geomOr(active poly))
removing unused task: pChannel = geomAnd(pNotOhmic poly)
removing unused task: nChannel = geomAnd(nNotOhmic poly)
removing unused task: dio_id = geomOr("dio_id")
removing unused task: cap_id = geomOr("cap_id")
removing unused task: nolpe = geomOr("nolpe")
removing unused task: bkgnd = geomBkgnd()
warn:   Duplicate check "(SCMOS Rule 20.9) sblock enclosure of poly resistor: 0.20 um" is being ignored.
warn:   Duplicate check "(SCMOS Rule 26.2) metal5 width: 0.30 um" is being ignored.
Running layout DRC analysis
Flat mode
Full checking.
DRC started.......Sat Apr 12 23:01:42 2025
    completed ....Sat Apr 12 23:01:45 2025
    CPU TIME = 00:00:00  TOTAL TIME = 00:00:03
*********   Summary of rule violations for cell "8bit_adder layout"   *********
   Total errors found: 0
```

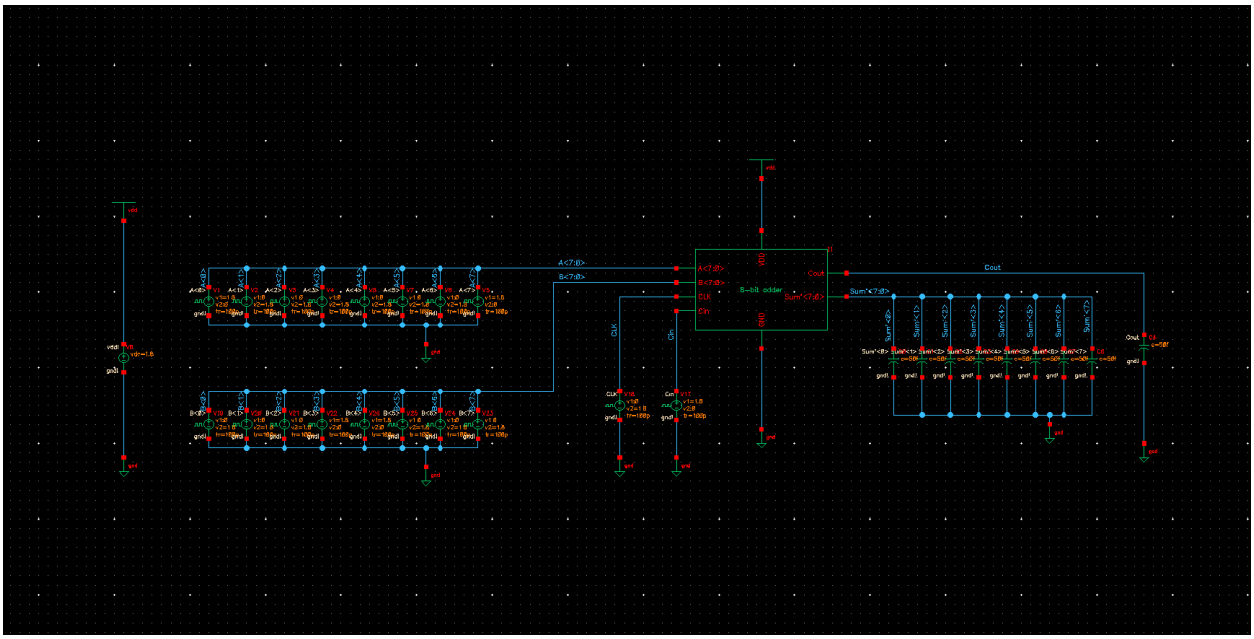
5. LVS:

```
@(#)$CDS: LVS version 6.1.8-64b 09/03/2024 18:56 (sjfhw316) $

Command line: /opt/coe/cadence/IC618/tools.lnx86/dfII/bin/64bit/LVS -dir /home/grads/y/yashas1729/LVS -l -s -t /home/grads/y/yashas1729/LVS/layout /home/grads/y/yashas1729/LVS/schematic
Like matching is enabled.
Net swapping is enabled.
Using terminal names as correspondence points.
Compiling Diva LVS rules...

    Net-list summary for /home/grads/y/yashas1729/LVS/layout/netlist
       count
        315             nets
        29              terminals
        305             pmos
        305             nmos

    Net-list summary for /home/grads/y/yashas1729/LVS/schematic/netlist
       count
        315             nets
        29              terminals
        305             pmos
        305             nmos


    Terminal correspondence points
    N297      N24       A<0>
    N293      N23       A<1>
    N289      N22       A<2>
    N286      N21       A<3>
    N312      N59       A<4>
    N308      N58       A<5>
    N303      N57       A<6>
    N301      N56       A<7>
    N314      N28       B<0>
    N310      N27       B<1>
    N306      N26       B<2>
    N302      N25       B<3>
    N299      N40       B<4>
    N296      N36       B<5>
    N292      N43       B<6>
    N288      N41       B<7>
    N295      N2        CLK
    N304      N29       Cin
    N307      N60       Cout
    N291      N1        GND
    N300      N38       Sum'<0>
    N298      N47       Sum'<1>
    N294      N39       Sum'<2>
    N290      N42       Sum'<3>
    N287      N55       Sum'<4>
    N313      N54       Sum'<5>
    N309      N53       Sum'<6>
    N305      N52       Sum'<7>
    N311      N0        VDD

Devices in the netlist but not in the rules:
        pcapacitor
Devices in the rules but not in the netlist:
        cap nfet pfet nmos4 pmos4

The net-lists match.

                              layout  schematic
                                 instances
        un-matched              0       0
        rewired                 0       0
        size errors             0       0
        pruned                  0       0
```

<u>Post Layout Simulations</u>

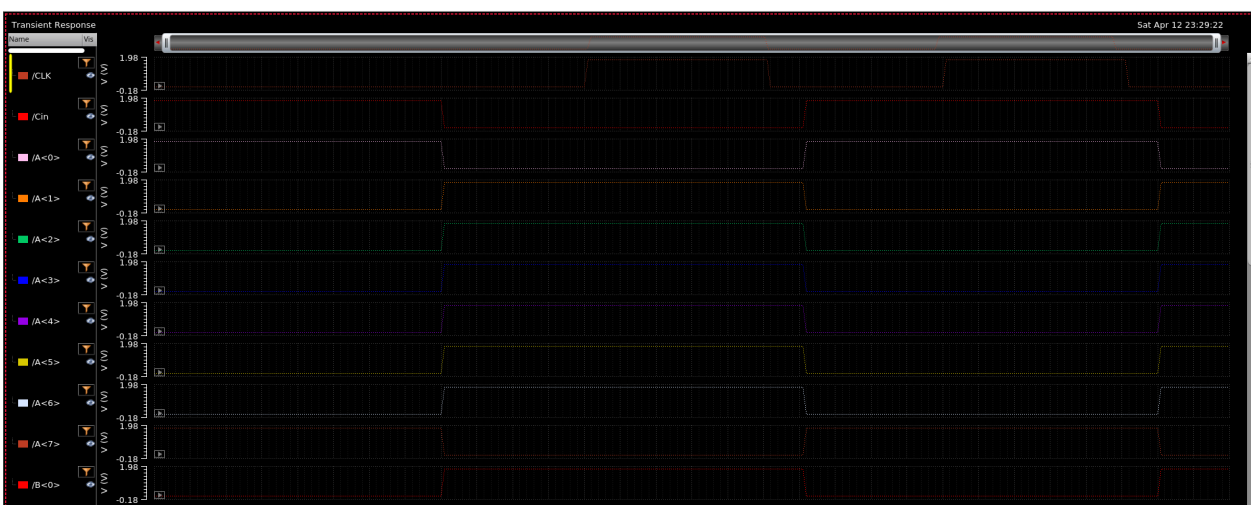
Simulation Setup:

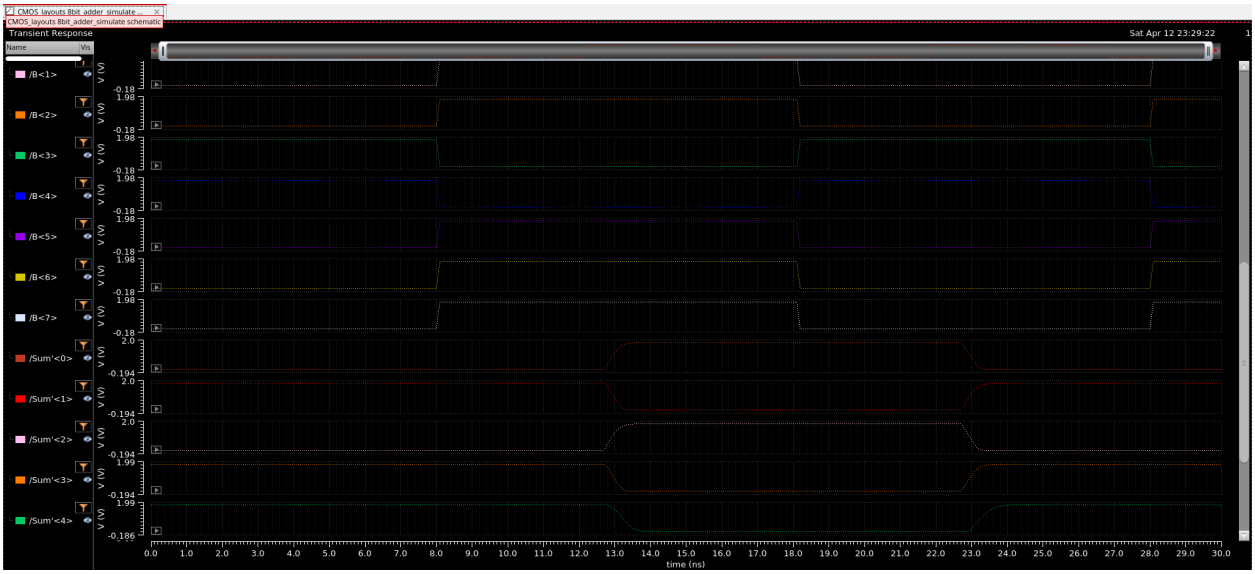
CLK:	Initial Delay = 12ns | Period = 10ns
A, B and Cin	Initial Delay = 8ns | Period = 20ns

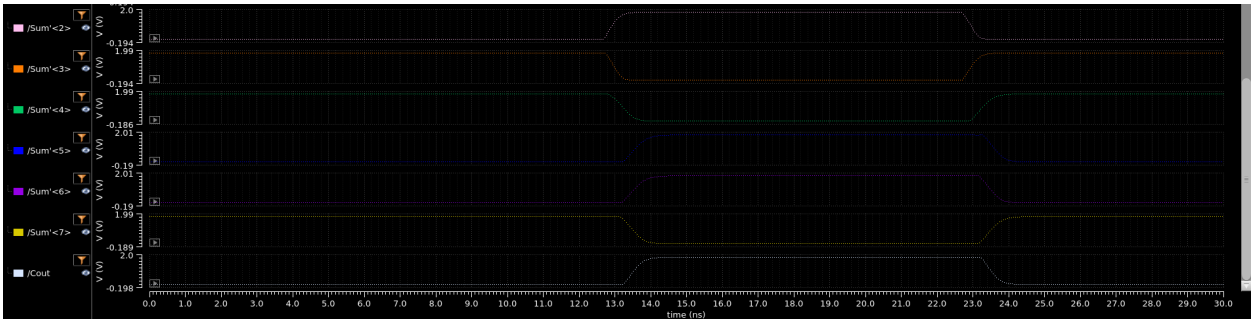


Case 1:

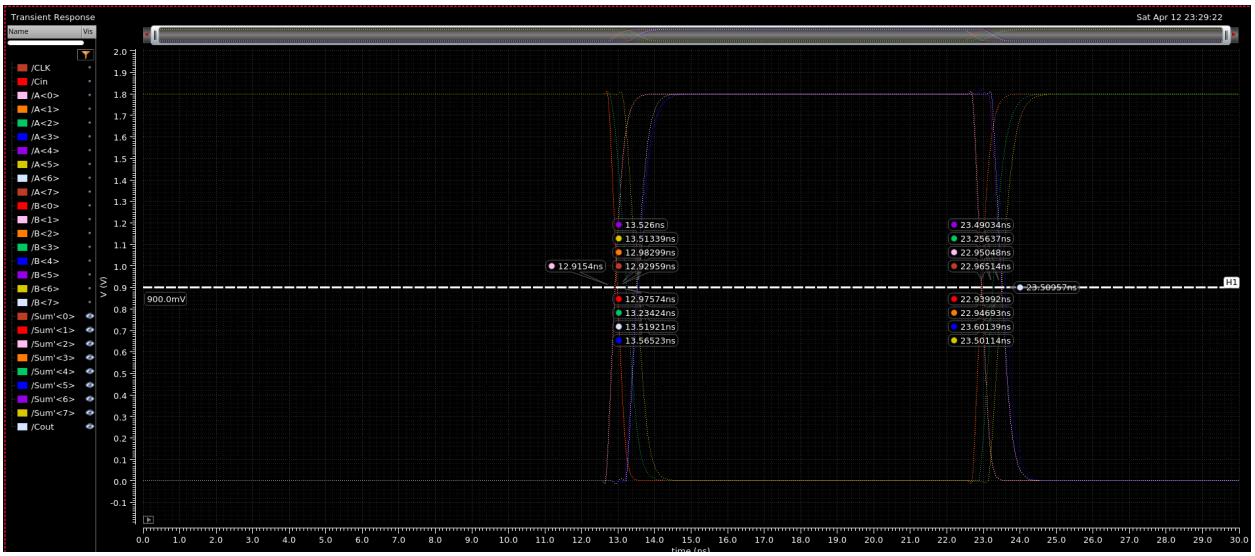
Input: A = 0111 1110 | B = 1110 0111 | Cin = 0
Output: Sum’ = 0110 0101 | Cout = 1







From the pictures above, we can see that the output matches the expected output.



As seen from the picture above, the maximum delay seen from the clock's rising edge transition to the output signal transition is for the Sum'<5> output.

t_d = 13.5652 - 12.05 = **1.5152 ns**

Additionally, the setup time that was calculated in lab 6 is:

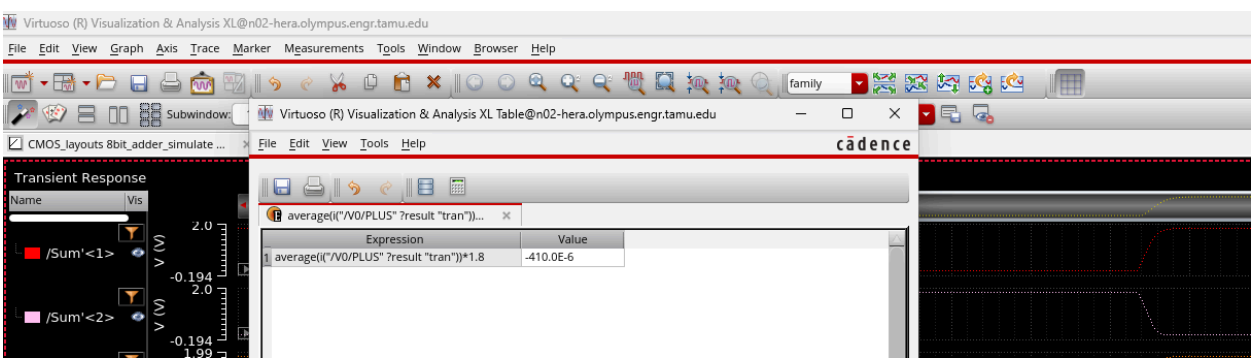
t_s = **0.182 ns**

Therefore, the clock period should be:

$t_{clk} > t_s + t_d$
Hence,
$t_{clk} > 1.515 + 0.182 \Rightarrow$ **t_{clk} > 1.697 ns**

Therefore, the maximum operating frequency = 589 MHz

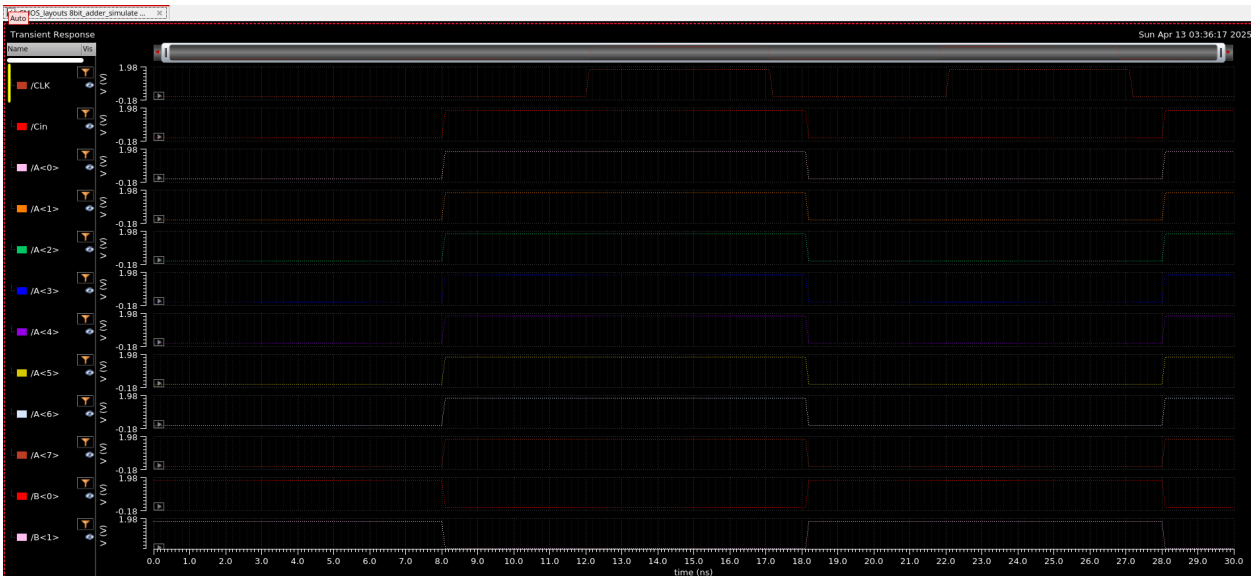
Power Consumed by VDD:

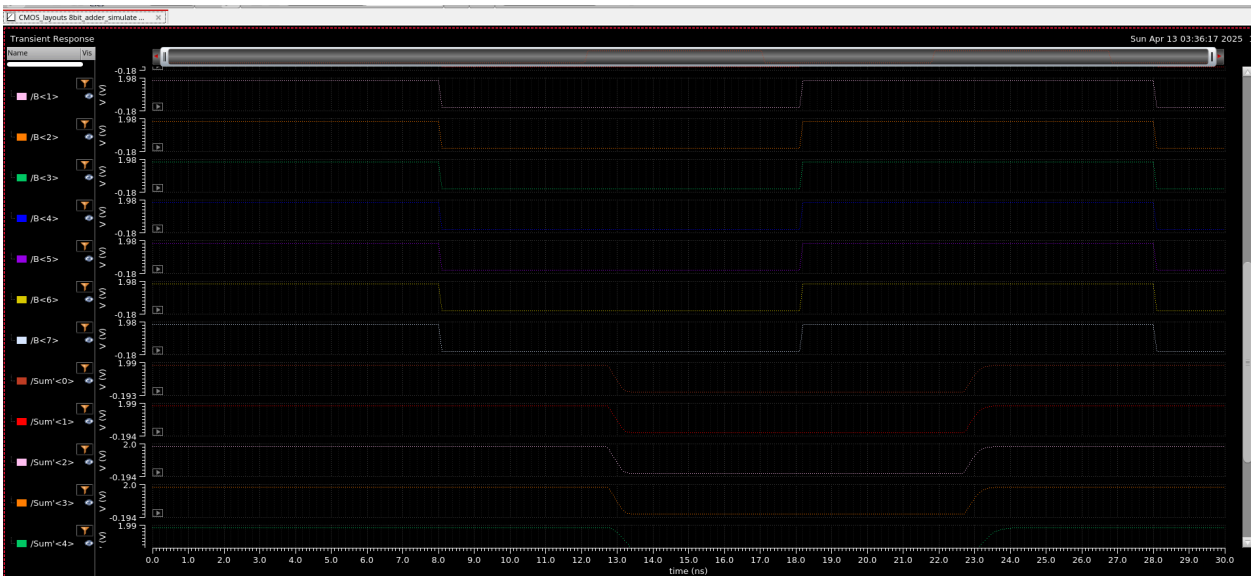


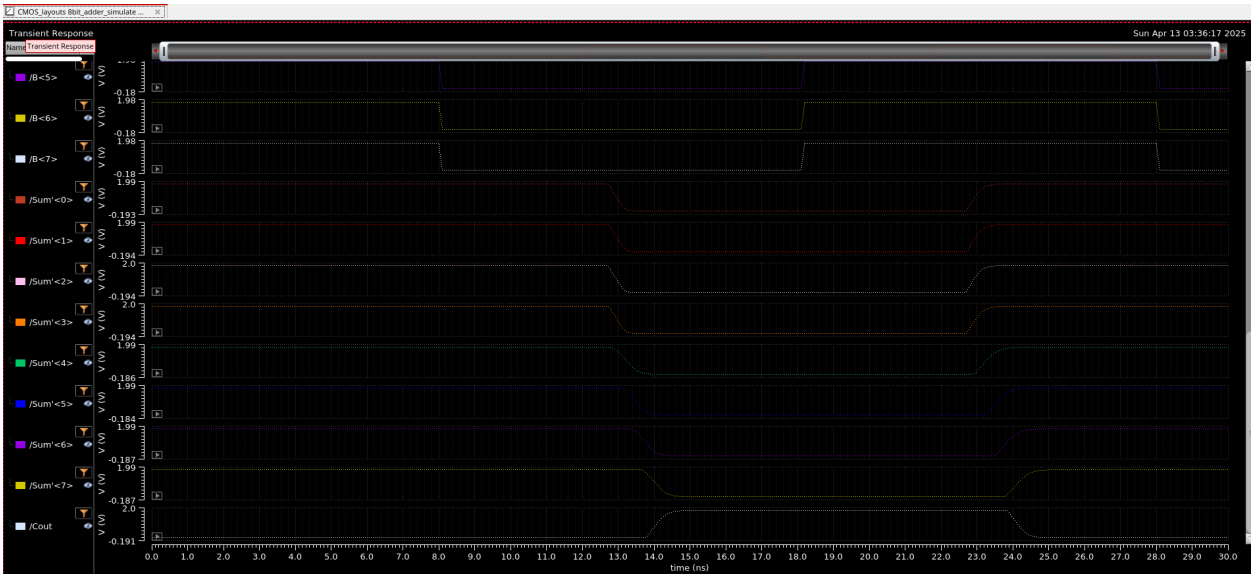
VDD Power = $410*10^{-6}$ W

Case 2:

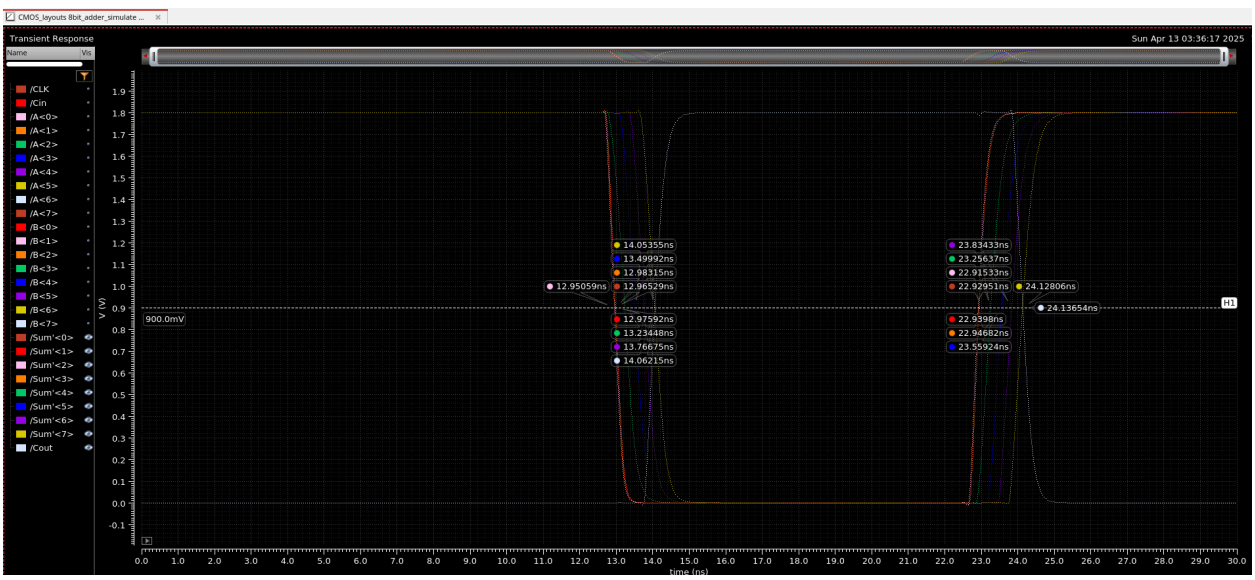
Input: A = 1111 1111 | B = 0000 0000 | Cin = 1
Output: Sum’ = 0000 0000 | Cout = 1







From the pictures above, we can see that the output matches the expected output.



As seen from the picture above, the maximum delay seen from the clock's rising edge transition to the output signal transition is for the Cout output.

t_d = 14.062 - 12.05 = **2.012 ns**

Additionally, the setup time that was calculated in lab 6 is:

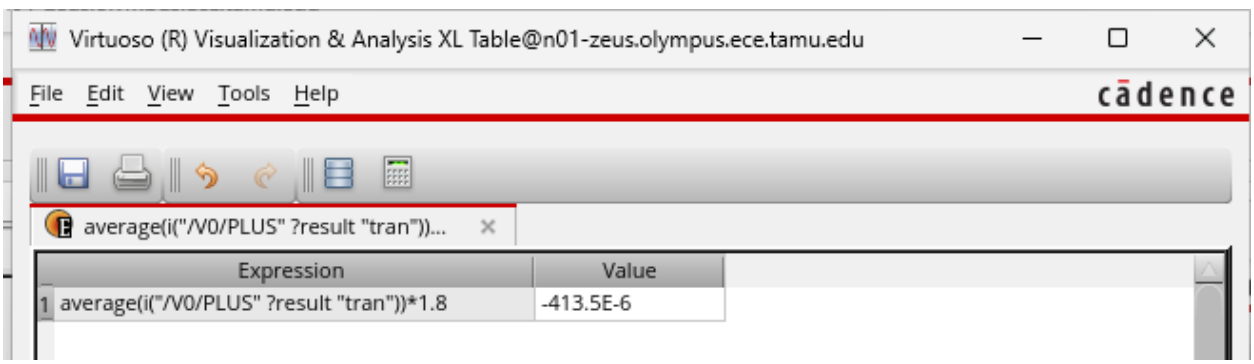
t_s = **0.182 ns**

Therefore, the clock period should be:

$t_{clk} > t_s + t_d$
Hence,
$t_{clk} > 2.012 + 0.182 \Rightarrow$ $\mathbf{t_{clk} > 2.194}$ **ns**

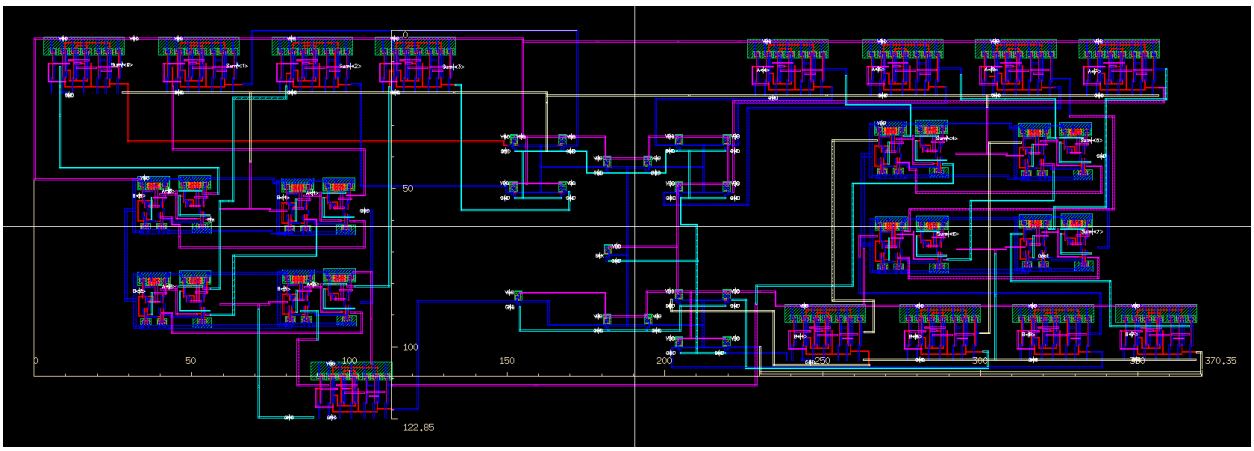
Therefore, the maximum operating frequency = 455 MHz

Power Consumed by VDD:



VDD Power = $413.5*10^{-6}$ W

Area Of Layout



Unit of the ruler: 10-6 m (micrometer)

Length = 370.35 x 10^{-6} m
Breadth = 122.85 x 10^{-6} m

Area = Length x Breadth = **4.549 x 10^{-8} m**